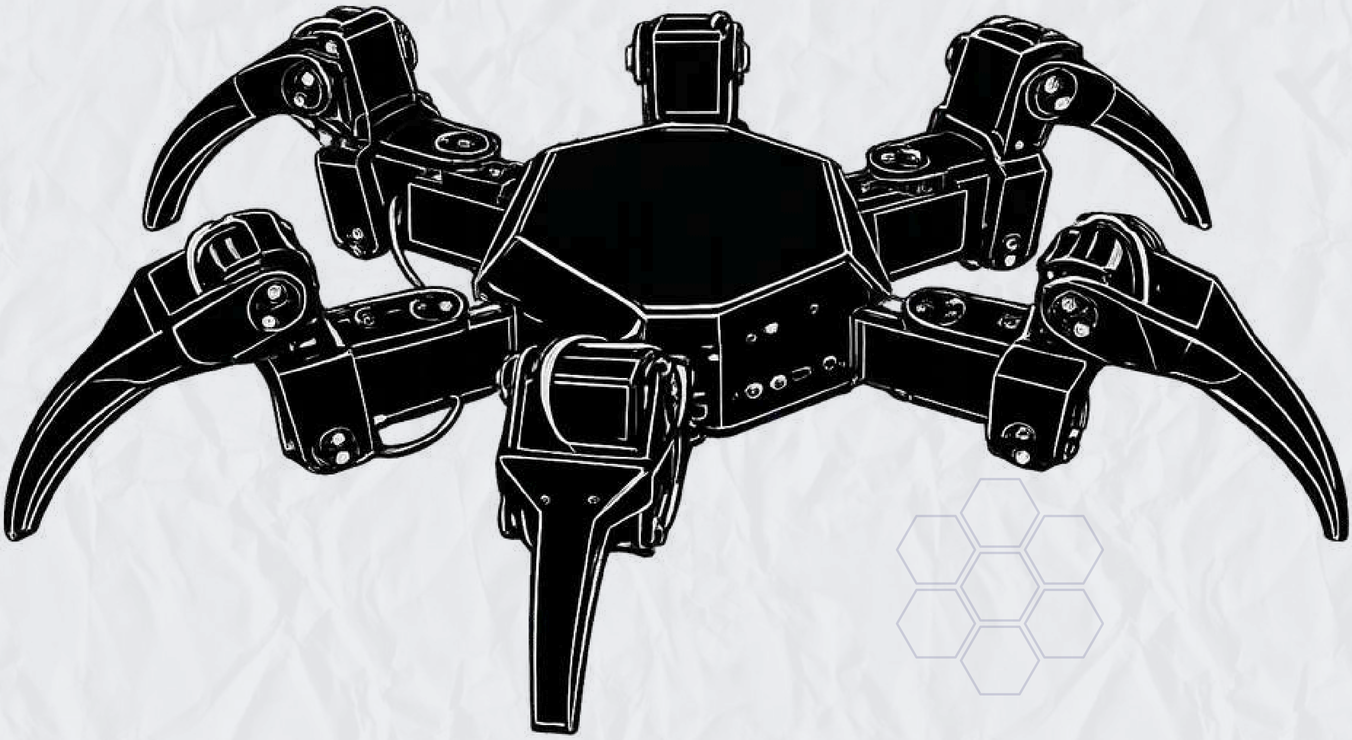
TEAM
SALIM ALI
BIO INSPIRED ROBOTICS

CM SHRI SCHOOL

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT



SENSE
Collects real-time environmental data

ANALYZE
Processes and interprets information

WARN
Identifies hazards and alerts in real time

PROTECT
supports safe decions and save lives

DEVELOPED BY

- ISHANT
- SARTHAK
- CHIRAG
- VIRAT
- AVIKA

UNDER GUIDANCE OF

DEVENDER PAWAR
&
OM MAHADESHWAR

Design build inteligence
for a safer tomorrow

THE INNOVATION STORY

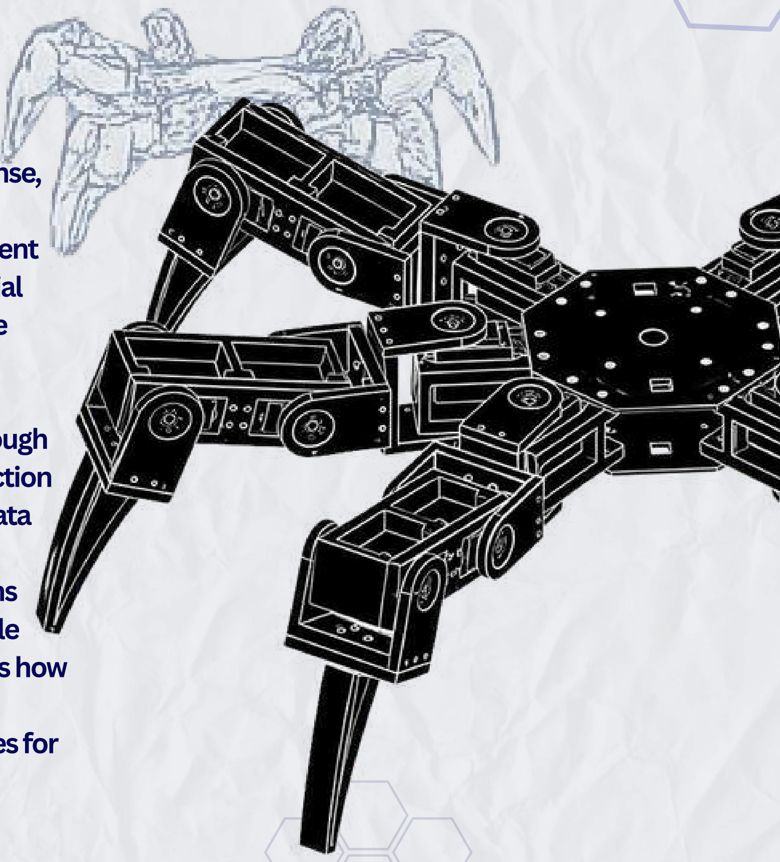
CENTRAL SQUARE FOUNDATION

LTM

SpyD
ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

PROJECT OVERVIEW



ABSTRACT

SpyD Rescue X6 is a spider-inspired hazard intelligence robot developed to enhance situational awareness in disaster response, industrial safety, and hazardous environment monitoring. The system combines multi-sensor environmental sensing, intelligent data analysis, and real-time risk assessment to identify potential threats and support informed decision-making. Inspired by the observational and adaptive behavior of spiders, the robot continuously monitors its surroundings, detects abnormal conditions, and generates actionable hazard information. Through sensor fusion, risk scoring, survivor localization, and SOS detection capabilities, SpyD Rescue X6 transforms raw environmental data into meaningful insights. Its modular architecture, wireless communication system, and autonomous monitoring functions enable effective deployment in challenging environments while reducing human exposure to danger. The project demonstrates how bio-inspired robotics can evolve beyond mobility to provide intelligent hazard assessment and decision-support capabilities for modern rescue and monitoring operations.

MISSION

To provide real-time hazard awareness, environmental intelligence, and decision support for rescue and monitoring operations.

VISION

To develop intelligent robotic systems capable of sensing, analyzing, and protecting lives in hazardous environments.

CORE OBJECTIVES

01 HAZARD DETECTION
Monitor environmental threats and abnormal conditions in real time.

02 INTELLIGENT ANALYSIS
Process and interpret sensor data through advanced sensor fusion techniques.

03 RISK ASSESSMENT
Evaluate, classify, and prioritize hazards based on severity and impact.

04 RESCUE SUPPORT
Assist rescue teams through survivor localization, SOS detection, and situational awareness.

SYSTEM WORKFLOW

MOVE
Navigate through challenging and hazardous environments.

SENSE
Collect real-time environmental data using multiple sensors.

ANALYZE
Fuse and process sensor information to extract meaningful insights.

RISK SCORE
RISK
Assign hazard levels ranging from Safe to Critical.

WARN
Generate alerts and notify operators about potential threats.

PROTECT
Support informed decision-making and improve operational safety.

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

MEET THE TEAM

Building Intelligence. Protecting Lives.

Ishant
Team Leader/ Mechnical Lead

- Project Planning
- CAD Modeling
- Assembly

Sarthak
Harware Lead

- Circuit Design
- Embedded Systems
- Power Managment

Chirag
Software Lead

- Data Communication
- Programming

Avika
UI/UX DESIGNER

- UI/UX Design
- Visual Documentation

Virat
Research and AI Lead

- AI & Data Analysis
- Research
- Digital Hazard Web

UNDER GUIDANCE OF

MR. DEVENDER PAWAR
TECHINCAL EXPERT

MR. OM MAHADESHWAR
MECHANICAL EXPERT

EXPERTISE
Robotics, embedded systems, AI, Sensor fusion, CAD designing and 3D modelling

ROLE
Mentoring, Review, Technical guidance, mechanical guidance & Problem solving

FOCUS
Mentoring, Review, Technical guidance , Mechanical guidance & Problem solving

PROJECT CONTRIBUTION

PLANNING
Defines goals, plan milestones and manage overall project timeline.

DESIGN
Mechnanical modeling, electronics design and system architecture.

DEVELOPMENT
Firmware coding, sensor integration and software development.

ANALYSIS
Data analysis, risk assessment and performance evaluation.

TESTNG
System testing, validation and real-world scenario verification.

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

PROBLEM STATEMENT

Disaster reponses and industrial safely operations involve environments that are unpredictable., unsatble , and extremely hazardous. Rescue teams often lack sufficient information about the conditions inside such environments, leading to **delayed decisions** and **increased risk to human lives**.

1. RESCUE CHALLENGE

Rescuers face numerous difficulties when operating in disaster or hazardous zones:

- collapsed structure and unstable terrain
- toxic gases, fire, smoke and low visibility
- narrow and collapsed spaces inaccessible to humans
- lack of real-time environmental information
- high risk of secondary collapse and human exposure

2. HAZARD AWARENESS GAP

In most rescue operations, teams enter unknown areas with limited or no prior information.

3. KEY CHALLENGES IDENTIFIES

UNKNOWN ENVIRONMENT
No prior information about internal conditions

HAZARDOUS CONDITIONS
Presence of toxic gases, heat, smoke and debris.

CONFINED SPACES
Inaccessible areas not suitable for humans or wheeled robots.

LACK OF INTELLIGENCE
Absence of real-time data and hazard interpretation.

HIGH HUMAN RISK
Rescuers exposed to life-thretening situations.

DELAYED RESPONSE
Slow assessment leads to delayed rescue actions.

4. OUR SOLUTION

SpyD Rescue X6 is a spider-inspired hazard intelligence robot designed to go where humans and traditional robots cannot.

1. Multi-sensor environmental sensing
2. hazard detection and monitoring
3. real-time risk assessment
4. Survivor & SOS detection
5. digital hazard web mapping
6. intelligent alerts & decision support

REAL-LIFE INSPIRATIONS

INCIDENTS THAT HIGHLIGHTS THE NEED FOR INTELLIGENT RESCUE ROBOTS

01 SILKAYA TUNNEL COLLAPSE

(UTTARAKHAND, 2023)

In November 2023, a section of the Silkaya Tunnel collapsed, trapping 41 workers inside the 17 days. Rescue teams faced extremely difficult conditions while trying to reach the trapped workers.

KEY CHALLENGES

NARROW & CONFINED SPACES
Very limited for humans entry and equipment movement.

UNSTABLE DEBRIS
High risk of further collapse and additional landslides.

LIMITED VISIBILTY
Dark, dusty and smoky environment restricted visibility.

DIFFICULTY IN ASSESSING CONDITIONS
Lack of real-time information bout the situation inside the tunnel

LESSON LEARNED

This incident highlighted the need for robotic system that can safely enter dangerous areas before human rescuers, provide real-time updates and reduce risks to human life.

02 BURARI BUILDING COLLAPSE

(DELHI,2019)

A muti-story building collapsed in Burari, Delhi, trapping several people beneath concrete rubble and damaged structures.

KEY CHALLENGES

UNSTABLE RUBBLE
Constant risk of secondary collapse while searching for survivors.

RISK OF FURTHER COLLAPSE
Rescuers had to work under highly unstable conditions.

LOCATING SURVIVORS
Finding survivors required careful inspection of confined spaces and extended efforts.

LESSON LEARNED

This incident demonsrates the need for a spider-inspired robot capable of operating in dangerous environments where human access in difficult and conventional robots may struggle.

WHY SPIDER-INSPIRED INTELLIGENCE?

LEARNING FROM NATURE. THINKING FOR RESCUE.

Spiders are master survivors. They sense their environment, detect vibrations, analyze threats, and adapt instantly to any situations. These natural abilities inspire SpyD Rescue X6 to become more than a robot that moves - it thinks, sense, analyzes and protects.

NATURE'S INTELLIGENCE

OBSERVE
Spiders continuosly observe their surroundings.

PERCIEVE
They percieve vibrations, airflow and subtle changes.

RESPONS
They make fast, intelligent decisions and act.

WHAT SPIDERS CAN DO

ENVIRONMENTAL AWARENESS
Spiders sense vibrations and detect minute changes in their environments.

ADAPTIVE MOVEMENT
They adapt their gait instantly to different terrains and surfaces..

OBSTACLE ASSESSMENT
Spiders evaluate obstacles before crossing and choose the safest path.

STABILITY ON COMPLEX TERRAIN
Multiple legs provide excellent stability even on unstable and eneven surfaces.

EFFICIENT BODY DISTRIBUTION
Weight is distributed across multiple legs for balance and control.

REAL-TIME DATA

SPIDER INTELLIGENCE + ADVANCED TECHNOLOGY = INTELLIGENT PROTECTION

SPIDER INSPIRATION → ROBOTICS INTELLIGENCE

- Vibration sensing → Multi-sensor system to detect environmental changes.
- Environmental Awareness → 360 degree situational awareness and hazard monitoring
- Adaptive behavior → Autonomous decision-making and path adaption
- Obstacle Evalution → Terrain analysis and safe navigation
- Multi-leg stability → Digital hazard web for mapping and the visualization
- Web as network → Digital hazard web for mapping and data visualization

NEED FOR HAZARD INTELLIGENCE SYSTEMS

AWARENESS TODAY. SAFETY TOMORROW

In dangerous and unknown environments, what we cannot see can harm us. Real-time hazard intelligence is essential to make faster, safer decisions and save more lives.

THE PROBLEM WITHOUT INTELLIGENCE

Unknown Environment → Limited Awareness → Delayed Decision → Higher Risk & Caualities

WHY INDIA NEEDS HAZARD INTELLIGENCE SYSTEMS

SILKAYA TUNNEL COLLAPSE(2023)

- 41 workers trapped for 17 days
- narrow dark and unstable tunnel condition
- robotics monitoring and situational awareness were highly needed.

BUILDING COLLAPSE INCIDENTS

- frequent building collapses in cities
- high risk of secondary collapse for rescuers.
- hazard mapping and structural assessment can greatly improve safety.

MINING & INDUSTRIAL ACCIDNETS

- gas leaks, toxic fumes and low visibility zones.
- difficult to access risk before human entry
- real-time environmental sensing can prevent disasters.

LANDSLIDES & EARTHQUAKES

- India is highly prone to natural disasters.
- rapid assessment saves time and resources
- early detection and survivors localization can save innumerable lives.

INDIA - A HIGH RISK NATION

- **58.6%** of the landmass is prone to earthquakes
- **40 MILLION+** people affected by floods every year
- **~15,000** landslides reported in the last decade
- **THOUSANDS** of industrial accident and gas leaks annually
- **100+** tunnel projects under construction across india

India faces frequent disasters, industrial accidents, tunnel collapses and infrastructure failures where real-time hazard awareness is often limited. SpyD Rescue X6 is designed to bridge this gap by providing environmental intelligence, risk assessment and decision-support capabilities before recue teams enter dangerous zones.

WHY INTELLIGENCE MAKES THE DIFFERENCE

- **SENSE** Continuously collects real -time data from the environment .
- **ANALYSE** Processes data to understand hazards.
- **ASSESS** Evaluates security and calculate risk scores.
- **ALERT** Sends instant alerts and guidance to operators.
- **PROTECT** Enables safer decisions and protects lives.

SpyD
ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

SPYD RESCUE X6 - OUR SOLUTION

A SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

BUILD TO SENSE, ANALYSE, ALERT, PROTECT

SpyD Rescue X6 is a spider-inspired robotic platform developed to explore how multi-legged robots can assist in future rescue and inspection operations. The prototype focuses on stable locomotion, wireless control, and visual monitoring through an onboard camera while providing a foundation for future hazard intelligence capabilities.

- Spider-inspired mobility
- Multi-sensor intelligence
- Real-time analysis
- Instant alerts & warnings
- Build to protect human lives

KEY FEATURES

6-Legged Adaptive Mobility
Spider-like locomotion for superior stability on rubble, slopes and irregualr terrain.

Wireless Robot Control
ESP32-based communication enables remote operation and maneuvering.

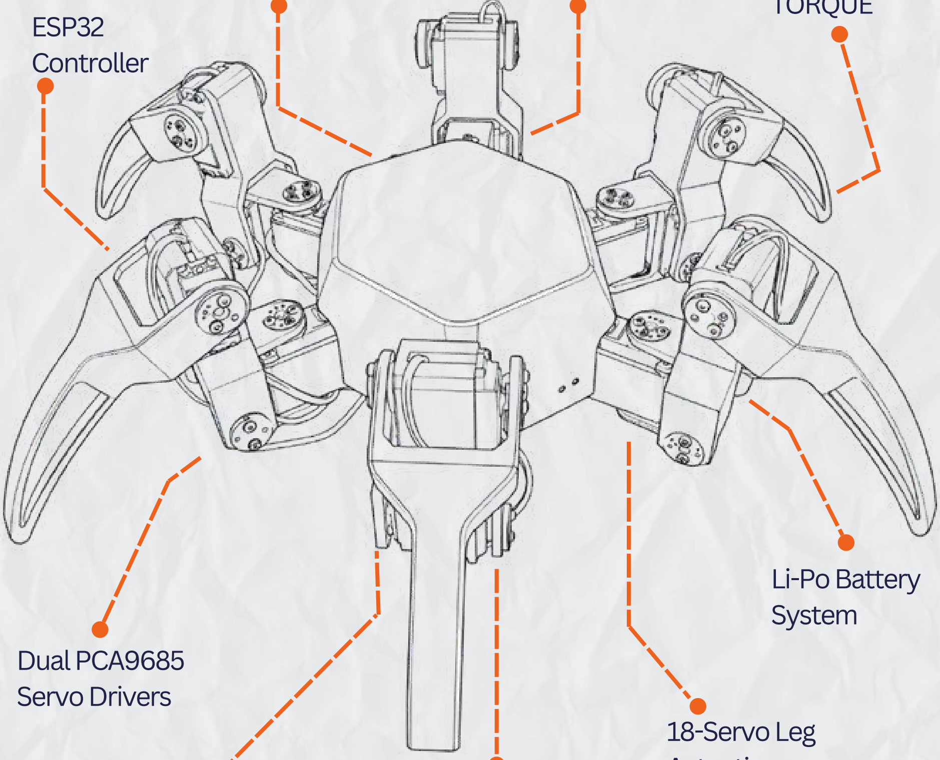
Modular Mechanical Design
The robot structure allows easy maintenance and future system expansion.

Camera-Based Monitoring
Provides live visual feedback for observation and navigation.

18-DOF Actuation
Eighteen servo motors enable coordinated leg movement and flexible motion.

Lightweight 3D Printed Structure
PLA-based construction reduces weight while maintaining strength.

Expandable Development Platform
Designed to support future integration of sensors and intelligent systems.



HOW IT WORKS

1. SENSE → 2. ANALYZE → 3. ALERT → 4. SUPPORT

TYPICAL USE CASES


Building Collapse


Tunnel Collapse


Industrial Accidents


Earthquake Zones


Mining Operations

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

PROJECT DEVELOPMENT OVERVIEW

A timeline of how ideas, research, and engineerign came together to buid SpyD Rescue X6

FROM CONCEPT TO CREATION

01

RESEARCH & STUDY

- Studies disaster scenarios
- Analysed existing robots
- Reviewed research papers
- Identifies key problem areas

02

SPIDER BEHAVIOR STUDY

- Observed spider locomotion
- Analysed satbility & adaptability
- Extracted key biological principles
- Applies to robot design

03

CONCEPT DESIGN

- Brainstormed multiple concepts
- Decided 6-legged structures
- Planned 3DOF per leg
- Finalized system approach

04

CAD MODELLING

- Design chassis & leg mechanism
- 3D modeling in Fusion 360
- Analysed weight & balance
- Optimized for strength & mobility

05

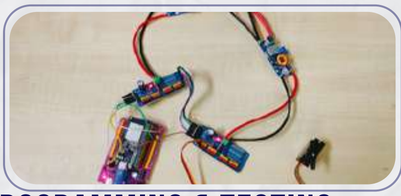
3D PRINTING

- Printed structural components
- Used PLA for durability
- Post-processed parts
- Quality checked each part

06

MECHANICAL ASSEMBLY

- Assembly body & legs
- Installed high torque servos
- Ensured proper alignment
- Tested movement & stability



07

ELECTRONICS INTEGRATION

- Integrated ESP32, PCA9685
- Connected all servos
- Designed power system
- Ensure secure wiring

08

PROGRAMMING & TESTING

- Developed movement alogrithms
- Integrated sensor data
- Tested all functions
- Validated in multiple conditions

KEY FOCUS AREAS

STABILITY

Build for rough terrain and unstable surfaces

RELIABILITY

Tested for durability and continuous operations

INTELLIGENCE

Real-time hazard detection and smart alerts

EFFICIENCY

Optimized power usage for longer missions

IMPACT

Designed to protect and save human lives.

PROJECT DEVELOPMENT JOURNEY

– OUR CONCEPT TO SPYD RESCUE X –

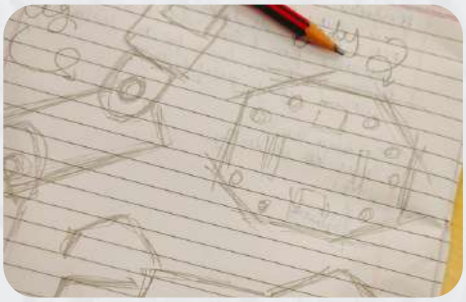
01 DAY 01-02
RESEARCH AND STUDY


Design of Six Legged Spider Robot and Evolving Walking Algorithms

Tolga Karakurt, Akif Durdu, and Nihat Yilmaz

- Extensive research on disaster scenarios and existing robots.
- Studied research papers, journals and case studies.
- Identified key problems in current rescue systems.

02 DAY 03-04
CONCEPT IDEATION



- Brainstormed multiple concepts.
- Decided on 6-legged structure with 3 DOF.
- Finalized major components and system approach.

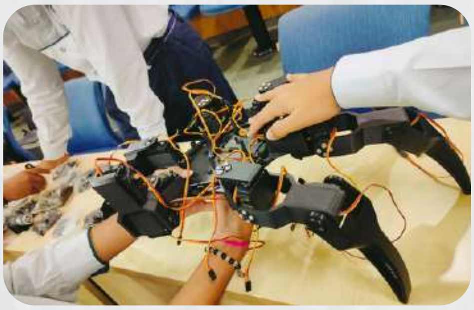
03 DAY 05-06
CAD MODELLING

- 3D modelling of chassis and legs in Fusion 360.
- Performed assembly simulation and motion study.
- Optimized design for strength and weight.

07 DAY 13-14
PROGRAMMING & WIREFRAME

- Developed movement algorithms for 6-legged locomotion.
- Programmed sensor data reading and processing.
- Implemented risk scoring and alert system .

08 DAY 15-16
GAIT TESTING



- Tested the walking of robot through different codes.
- Make the robot sit, stand and walk on normal plain surface.
- Change the speed of the robot with time.

09 DAY 17-18
SYSTEM INTEGRATION

- Integarted hardware and software full system.
- Validated data flow from sensors to dashboard.
- Ensured real-time responses and communication.

KEY DECISIONS

- 6-legged structure for maximum stability
- 3DOF per leg for natural movement
- Multi-sensor fusion for reliable data.
- Real-time risk scoring for quick decisions.

CHALLENGES FACED

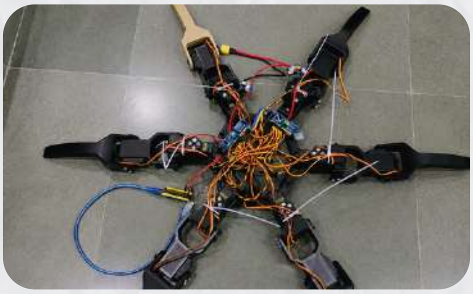
- Servo overheating during movement
- Power fluctuations affecting sensors.
- Sensor interference in confined spaces.
- Balancing weight and stability

"A journey of ideas, Learning challenges and engineering solution."

04 DAY 07-08 3D PRINTING

- Printed all structural parts with high infill for durability .
- Post-processing and fitting of printed componenets .
- Prepared mechanical parts for assembly.

05 DAY 09-10 MECHANICAL ASSEMBLY



- Assembly chassis and all six legs.
- Installed and alignmed high torque servos.
- Tested leg movement and stability.

06 DAY 11-12 ELECTRONIC INTEGRATION

- Integrated ESP32, PCA9685, and power system.
- Connected sensors and communication modules.
- Ensured proper power distribution and safety .

04 DAY 19-20 TESTING & VALIDATION

- Performed movement testa on various terrains.
- Validated hazard detection and alert generation.
- Checked stability, endurance and reliability.

05 DAY 21-22 SYPD RESCUE X6

- Final system ready for real-world scenarios.
- Hazard intelligence, real-time alerts, saving lives.
- Mission ready. Always ready

HOW WE OVERCOME

- Added cooling & optimized movement
- Designed dedicated power modules.
- Calibrated sensors & filtering algorithms
- Lightweight design with strong structure

THIS JOURNEY TAUGHT US....

"Innovation is not just about building something new, it's about solving real problems with dedication teamwork and engineering."

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

MECHANICAL SYSTEM DESIGN

SIMPLE, STRONG, ADAPTIVE

SpyD Rescue X6 is designed with a minimal yet robust mechanical structure inspired by the movement of spiders. It is built to crawl, climb and adapt to rough and unpredictable terrains.

DESIGN PHILOSOPHY

STABILITY
Low center of gravity for maximum balance on uneven terrain.

SIMPLICITY
Minimal components for reliability and easy maintenance.

ADAPTABILITY
Spider-inspired 6-leg mechanism for climbing, crawling and gripping.

DURABILITY
Lightweight structure with strong materials for rough environments.

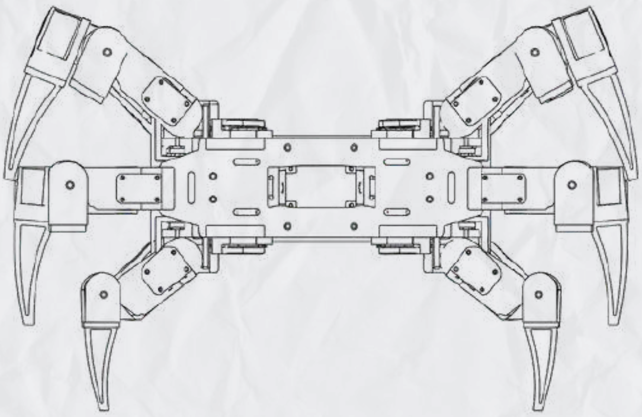
MECHANICAL OVERVIEW

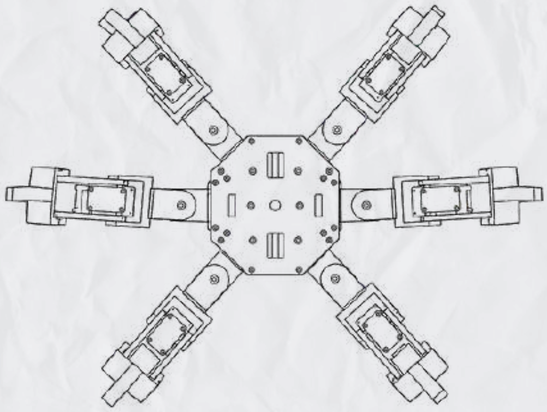
6-LEG LOCOMOTION
For stability and obstacle negotiation.

COMPACT BODY
Low-profile chassis to fit in narrow spaces.

HIGH GROUND CLEARANCE
Allows movement over rocks, debris and edges.

MODULAR LEG DESIGN
Independent multi-joint legs improve maneuverability and simplify maintenance.



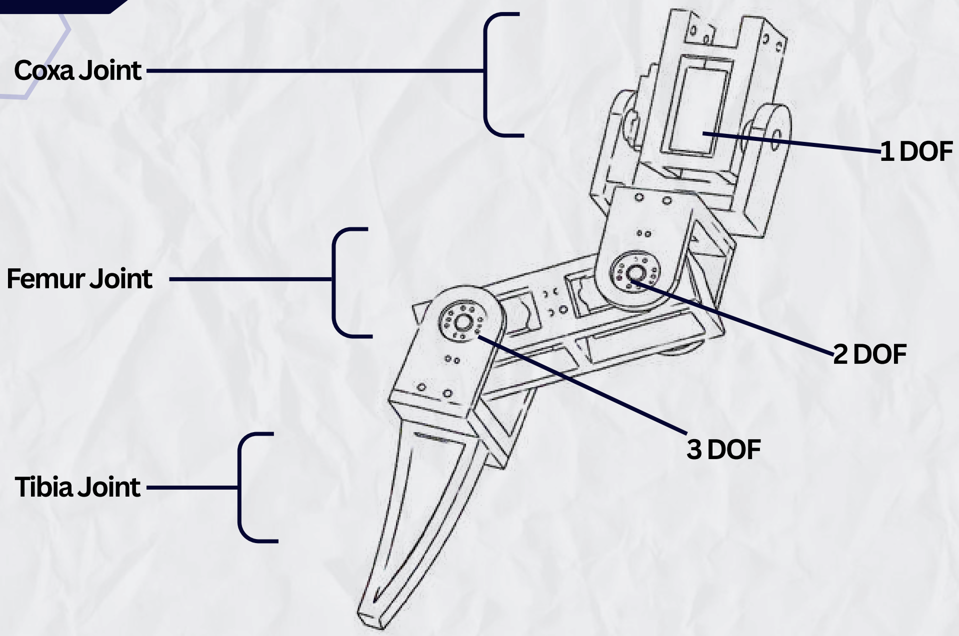


STRUCTURAL LAYOUT

- Inspired by spider biomechanics for natural and efficient movement .
- Minimum components, maximum performances.
- Built for real-world rescue environments.
- Modular design for easy repairs and upgrades.

> *"Every joint, every link and every movement is designed with a single goal – to reach where others can't and save where it matters most."*

LEG MECHANISM



MATERIALS AND COMPONENTS

COMPONENTS	MATERIAL/PARTS
MG995 Servo Motors(18x)	Actuation of all leg joints
PLA 3D Printed Parts	Chassis and Leg structure fabrication
Hexagonal Central Body Plates	Main structural frame
Servo horns & Mounting Brackets	Motion transfer and joint assembly
Nuts, Bolts & Spacers	Mechanical fastening
ESP32 Development Board	Main controller
PCA9685 Servo Driver Modules(2x)	Control of multiple servos
Buck Converters(2x)	Voltage regulation
LiPo Battery	Power Source
PS2 Joystick Module	Manaul wireless control
Camera Module	Real-time visual monitoring

KEY DIMENSIONS

PARAMETERS	VALUE
Number of Legs	6
Servo Motors	18(3 per leg)
Degrees of Freedom	18 DOF
Body Shape	Hexagonal
Leg Configuration	3-Joint Articulated Leg
Chassis Type	Central Hexapod Frame
Frame Material	PLA (3D printed)
Camera Moutn Position	Front Center
Weight	3.5 kg (approx.)

SpyD ResQ X6

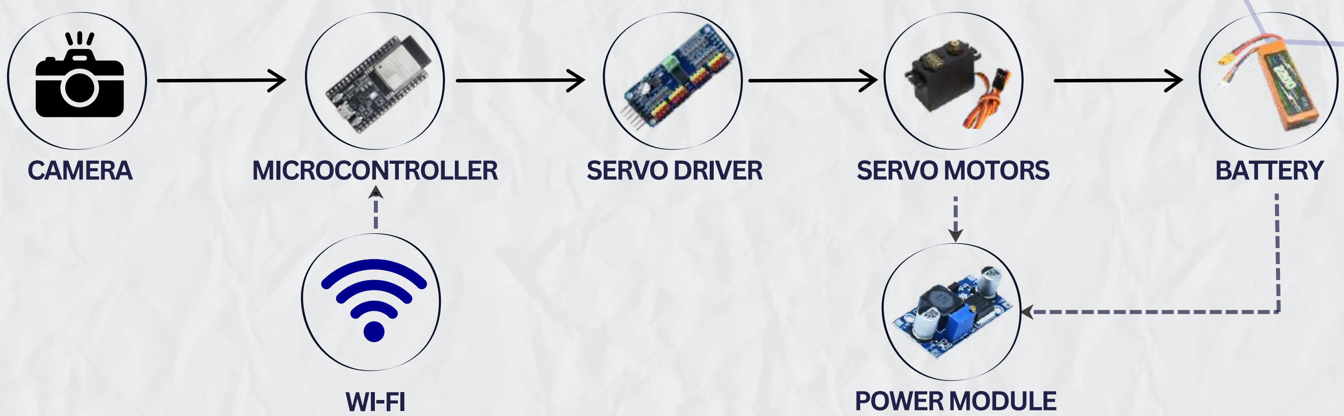
SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

ELECTRONICS INTEGRATION

S M A R T , E F F I C I E N T , R E L I A B L E

SpyD Rescue X6 uses a simple and efficient electronics system focused on reliable communication, real-time video and smooth movement. Only the essential componenets are used to keep the robot lightweight , power-efficient and mission-only.

MATERIALS AND COMPONENTS

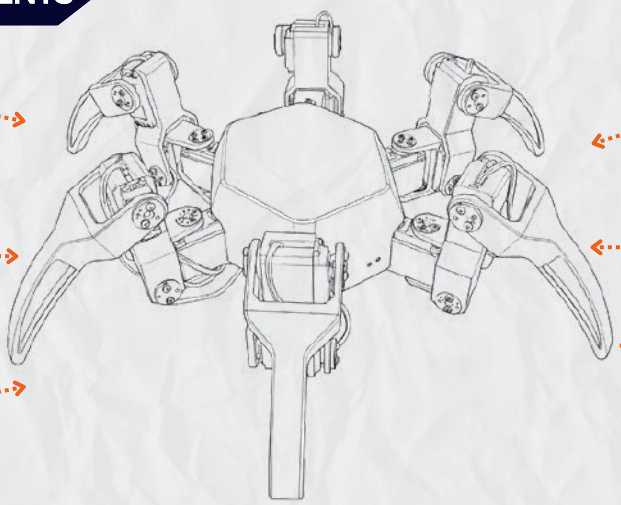


MATERIALS AND COMPONENTS

PLA for 3D printing

ESP32 DEVKIT

PCA9685 SERVO DRIVER



SERVO MOTORS (18)

POWER MODULE (5V/6V)

BATTERY

The SpyD Rescue X6 electronic system combines an ESP32 controller, dual PCA9685 servo drivers, and a regulated power network to ensure reliable operation. An onboard camera provides real-time visual monitoring, while the organized wiring architecture simplifies integration, testing, and future upgrades.

WHY THIS SETUP ?

Minimal & Essential Components

Lightweight & Power Efficient

Real-Time Communication

Reliable & Easy to Maintain

COMPONENETS USED

1. PCA9685 SERVO DRIVER

- Controls up to 16 servos at the same time.
- Uses only two pins to talk to your microcontroller.
- Has high precision with 4,096 different position steps.

2. LiPo BATTERY

- Provides a strong 11.1 volts of total electrical power.
- Stores 3300 milliamp-hours of total energy for runtime capacity.
- Delivers up to 115.5 amps of continuous current safely.
-

3. MG995

- Rotates between 0 and 180 degrees using standard PWM signals.
- Delivers high torque up to 10 kilograms of pulling force.
- Moves quickly with a top speed of 0.16 seconds per 60 degrees.

4. PS2 JOYSTICK MODULE

- Uses two potentiometers to track movement along the X and Y axes.
- Converts directional joystick tilt into readable analog voltage signals.
- Outputs a central voltage value when the joystick sits at rest.

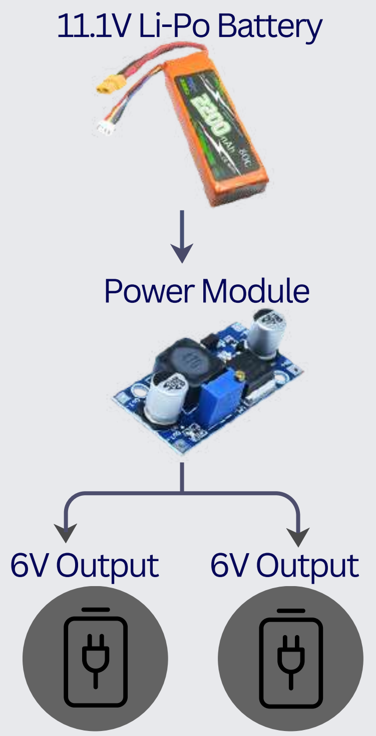
5. POWER MODULE (BUCK CONVERTOR)

- Regulates voltage for safe operation
- Acts as a step-down voltage regulator to lower input voltage safely.
- Converts your high-voltage LiPo power into a lower, stable voltage output.

6. ESP32

- Features a fast dual-core processor running at up to 240 MHz.
- Includes built-in Wi-Fi and Bluetooth for wireless communication projects.
- Provides a large number of digital and analog input/output pins.

POWER DISTRIBUTION



Efficient power usage for a long time

Over-voltage and reverse polarity protection

Stable performance in all conditions

KEY BENEFITS

- Real-time situational Awareness
- Smooth & Prescise Movements
- Low power consumption
- Simple, Reliable & Robust
- Built for rescue missions

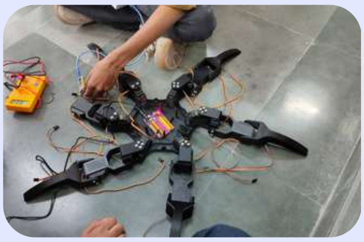
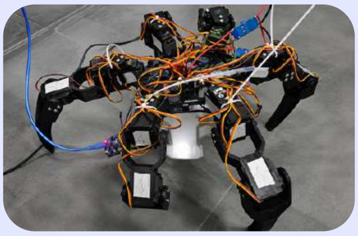
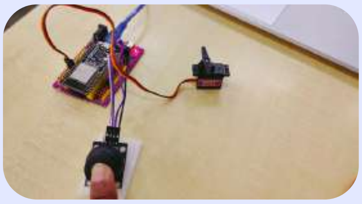
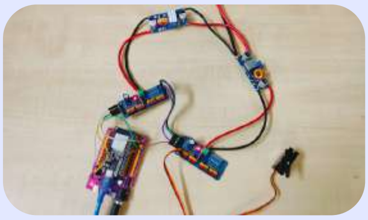
SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT

ENGINEERING CHALLENGES & LESSON LEARNED

S M A R T , E F F I C I E N T , R E L I A B L E

Every challege helped us learn, improve and build a more reliable robot

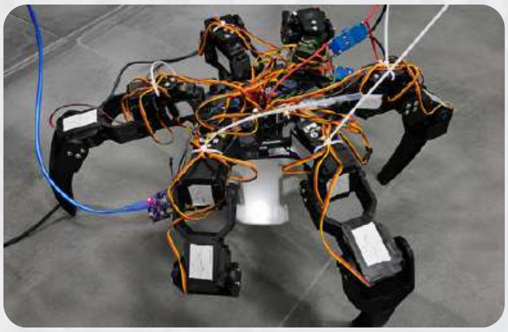
CHALLENGES	INVESTIGATION	SOLUTION IMPLEMENTED	IMAGES
01 SERVO CALIBRATION & ALIGNMENT Robot could not stand properly. Servos were not perfectly aligned to 90° neutral position.	• Checked servo horns and joints orientations. • Compared actual positions with required neutral orientation.	• Recaliberated all 18 servos to exact 90 degree usinf ESP32. • Adjusted servo horns and mounting positions.	
02 STANDING STABILITY Walking code was running, but robot collapsed when placed on ground.	• Obstacle posture and leg angles. • Tested by lifting robot on a box to remoce load.	• Corrected servo alignment. • Recaliberated neutral positions. • Adjusted initial leg angles in code.	
03 INCONSISTENT SERVO RESPONSE Some legs did not respond after power-up occasionally.	• Checked servo connections. • Verified power supply lines. • Monitored ESP32 communication.	• Secured loose connections. • Improved wiring management. • Ensured stable power distribution.	
04 X-Y JOYSTICK MAPPING ERRORS Servo move ment did not match joystick directions.	• Monitored analog values X and Y. • Compared output with expected movements.	• Debugged program logic. • Corrected X and Y value mapping. • Retested and calibrated .	
05 MULTI-SERVO SYNCHRONIZATION Coordinating 18 servos simultaenously was difficult .	• Tested with 3-leg subsystem first . • Observed timing and sequence differences.	• Used dual PCA9685 modules. • Organized channel assignment. • Gradually scaled to all 18 servos.	
06 WIRING & POWER MANAGEMENT Servo number of servo required organized wiring and stabel power distribution.	• Analyzed current requirement. • Identified need for regulated power. • Checked voltage drops.	• Used 2 bucks convertors. • Proper power routing. • Separated power and signal lines.	

RESULT & FUTURE SCOPE

Built to make a difference. Ready for the future.

Extensive testing and real-world trials proved that SpyD Rescue X6 is a reliable, agile, and efficient robot for search, surveillance, and rescue missions in hazardous environments.

TESTING RESULTS



TERRAIN ADAPTABILITY

Basic leg movement was tested on a raised platform. Servos responded correctly and followed the programmed gait sequence.

LEG COORDINATION

All servo motors were individually tested and successfully executed the programmed movement sequence.

POWER SYSTEM

Battery and power distribution were tested successfully for servo operation during laboratory testing.

STRUCTURAL STABILITY

The frame and leg assembly remained mechanically stable during movement trials.

BATTERY PERFORMANCE

Operates up to 45 minutes on a single charge under normal conditions.

ACHIEVEMENTS

- Designed and built a fully functional six-legged spider-inspired rescue robot.
- Lightweight, compact and robust structure for extreme terrains.
- Integrated multiple servo motors into the leg system. Successfully tested individual servo operation.
- Smooth and stable movement with high-torque digital servos.
- Efficient power management for longer mission time.
- Established a foundation for future locomotion testing and development.

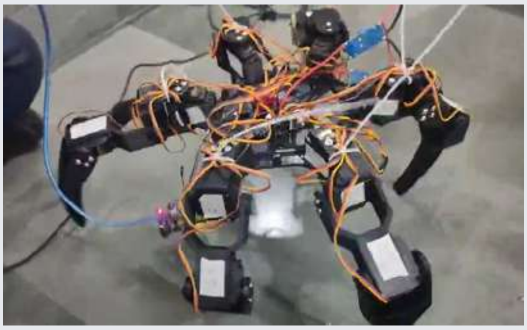
> **"Technology is the most powerful when it protects humanity."**
>
> **SpyD ResQ X6**

FUTURE SCOPE AND UPGRADES

- Integrate AI to detect humans, obstacles, fire, and hazards autonomously.
- Adding thermal vision to detect heat signatures in smoke, darkness, and rubble.
- Include gas, temperature, humidity, and smoke sensors for earliest hazard detection.
- Enable 3D mapping, obstacle avoidance, and autonomous path planning.
- Implement autonomous navigation and remote control with motion planning.

SpyD ResQ X6

SPIDER-INSPIRED HAZARD INTELLIGENCE ROBOT







The journey of developing SpyD Rescue X6 has been a valuable experience in engineering design, problem-solving, and innovation. From initial research and concept development to assembly, testing, and system integration, every stage contributed to a deeper understanding of robotics and intelligent rescue technologies. While the current prototype establishes a strong foundation, it also opens opportunities for future advancements in autonomous navigation, hazard detection, and disaster-response applications. This project reflects our commitment to learning, creativity, and the pursuit of technology that can contribute to safer and more effective rescue operations.